

Solving Equations & Inequalities

Module 3 Test

Grade 9 • MTH1W

Time: 60 minutes • Total: 70 marks

Name: _____

Date: _____

Instructions: Show all work for full marks. Marks for each question are shown in brackets. Calculators are permitted unless your teacher states otherwise.

Section A — Knowledge & Understanding

1. Solve: $x + 6 = 13$. **[1]**

2. Solve: $x - 8 = 5$. **[1]**

3. Solve: $4x = 32$. **[1]**

4. Solve: $x/2 = 9$. **[1]**

5. Solve: $2x + 1 = 11$. **[1]**

6. Solve: $5x - 3 = 17$. **[1]**

7. Solve: $x + 3 > 7$. **[1]**

8. Solve: $y - 5 = -2$. **[1]**

9. Solve: $6x = 42$. [1]

10. Solve: $4x + 3 = 27$. [2]

11. Solve: $5x - 7 = 18$. [2]

12. Solve: $x/4 + 2 = 5$. [2]

13. Solve: $2(x + 5) = 16$. [2]

14. Solve: $7x + 2 = 3x + 14$. [2]

15. Solve and describe the graph: $2x \leq 10$. [2]

Section B — Application

16. Solve: $6x - 5 = 2x + 15$. [2]

17. Solve: $2x/3 + 4 = 10$. [2]

18. Solve: $3(x - 4) = 2x + 1$. **[2]**

19. Solve $V = lwh$ for h . **[2]**

20. Solve: $5 - 3x = -7$. **[2]**

21. Solve and describe the graph: $3x - 2 > 7$. **[2]**

22. Solve: $4(x + 2) = 2(x + 9)$. **[3]**

23. Solve $A = (1/2)bh$ for b . **[3]**

24. Solve: $x/2 + x/3 = 5$. **[3]**

Section C — Thinking

25. The sum of three consecutive integers is 51. Find the integers. **[4]**

26. Solve and describe the graph: $-4x + 3 \geq 19$. **[4]**

27. The formula $F = (9/5)C + 32$ converts Celsius to Fahrenheit. Solve the formula for C, then find C when $F = 212$. (Very hard.) **[5]**

28. Solve: $(x + 2)/3 - (x - 1)/4 = 2$. (Very hard.) **[5]**

29. A rectangle's length is 3 cm more than twice its width. Its perimeter is 48 cm. Find the width and the length. (Very hard.) **[6]**

Section D — Communication

30. Explain why solving $-2x > 6$ requires reversing the inequality sign. Give the solution and describe its graph. (Very hard.) **[4]**

— END OF EXAM —